

Program: Diploma in Automation and Robotics	
Course Code:	Course Title: Mechanics of materials Lab
Semester: 3	Credits: 1.5
Course Category: Program core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 30

Course Objectives:

- To understand mechanical behavior of various engineering materials by conducting standard tests.
- To understand the procedure of doing different tests like tension, compression, hardness, torsion and impact in various materials
- To impart knowledge about the testing of springs.

Course Prerequisites:

Topic	Course code	Course name	Semester
Strength of materials	RAPC203	Mechanics of materials	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	The study the behavior of engineering materials under different loading conditions	6	Applying
CO2	Estimation of Hardness, Impact and toughness properties of engineering materials	9	Applying
CO3	Conduct torsion test on materials	3	Applying
CO4	Conduct spring test on materials	6	Applying
	Series Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline (Perform minimum Eight Experiments)

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	The study the behavior of engineering materials under different loading conditions		
M1.01	To study the behavior of mild steel under tension and to find out the young's modulus, ultimate strength, yield strength, percentage elongation.	3	Applying
M1.02	To find out the maximum shear stress in the mild steel specimen using double shear method.	3	Applying
Experiments Tension test on mild steel - To study the behavior of mild steel under tension and to find out the young's modulus, ultimate strength, yield strength, percentage elongation. Double shear test on mild steel - To find out the maximum shear stress in the mild steel specimen using double shear method.			
CO2	Estimation of Hardness, impact and toughness properties of engineering materials		
M2.01	Study on estimation of Hardness properties of engineering materials	3	Applying
M2.02	Study on estimation of Impact properties of engineering materials	3	Applying
M2.02	Study on estimation of Toughness properties of engineering materials	3	Applying

Experiments			
Brinell hardness Test- Study on estimation of Hardness properties of engineering materials by performing Brinell hardness test. Rockwell hardness Test - Study on estimation of Hardness properties of engineering materials by performing Rockwell hardness test. Vicker's Hardness Test -Study on estimation of Hardness properties of engineering materialsby performing Vicker'shardness test. Charpy test – Study on estimation of impact property of the engineering materials by performing Charpy test. Izod test – Study on estimation of impact property of the engineering materials by performing Izod test. Toughness test - To study the toughness property of materials.			
	Series Test – I	3	
CO3	Conduct torsion test on materials		
M3.01	To determine the modulus of rigidity the material of the given body	3	Applying
Experiments			
Torsion test – To determine the modulus of rigidity of the material of the given body			
CO4	Conduct a spring test on materials		
M4.01	To find the modulus of rigidity of the spring using compression test	3	Applying
M4.02	To find the modulus of rigidity of the spring using tension test	3	Applying
Experiments			
Compression test on spring - To find the modulus of rigidity of the spring using compression test Tension test on spring - To find the modulus of rigidity of the spring using tension test			
	Series Test– 2	3	

Text / Reference

T/R	Book Title/Author
R1	G E Dieter. Mechanical Metallurgy, McGraw Hill,2013
R2	S.S Rattan, “Strength of Materials”, McGraw Hill, 2nd edition, 2011.
R3	SS Bhavikatti ,Strength of materials –Vikas Publishing House